1 **(a)** All three points A, B, and C plotted accurately on the grid **[B1]**

(b) Identifies the horizontal and vertical translation, e.g. $\vec{BA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ OR correctly plots D on the grid **[M1]**

$(3, 5)$ **[A1]**

(c) Calculates the midpoint of AC or BD: $\left(\frac{-1+6}{2}, \frac{3+1}{2}\right)$ **[M1]**

$(2.5, 2)$ **[A1]**

[Total Marks: 5]

2 **(a)** Identifies the area of the enclosing square as a^2 and the three outer right-angled triangles as $\frac{1}{2}ab$, $\frac{1}{2}ab$, and $\frac{1}{2}(a-b)^2$ **[M1]**

Sets up the subtraction for the area of OPQ: Area $= a^2 - \frac{1}{2}ab - \frac{1}{2}ab - \frac{1}{2}(a-b)^2$ **[M1]**

Correctly expands brackets and simplifies algebraically to $\frac{1}{2}(a^2 - b^2)$ without errors **[A1]**

(b) Sets up the distance squared equation: $(a-b)^2 + (b-a)^2 = (4\sqrt{2})^2$

2(a-b)$^2 = 32$ [M1]

Solves to find $a - b = 4$ (accepting $a > b$) **[A1]**

Uses the area equation:

$$\frac{1}{2}(a^2 - b^2) = 24$$

a$^2 - b^2 = 48$ [B1]

Factorises to $(a-b)(a+b) = 48$, substitutes $a - b = 4$ to obtain $a + b = 12$ **[M1]**

$a = 8$ and $b = 4$ **[A1]**

[Total Marks: 8]

3 **(a)** Substitutes $y = 0$ to find $A(6, 0)$ **[B1]**

Substitutes $x = 0$ to find $B(0, 9)$ **[B1]**

(b) Recognises the change in x is -6 and change in y is 9 from A to B **[M1]**

Forms a valid ratio equation for either coordinate, e.g. x-coordinate $= 6 + \frac{2}{3}(-6)$ OR y-coordinate $= 0 + \frac{2}{3}(9)$ **[M1]**

$(2, 6)$ **[A1]**

(c) Substitutes $x = 4$ and $y = 3$ into the equation: $3(4) + 2(3)$ **[M1]**

Shows $12 + 6 = 18$ and clearly concludes 'Yes' **[A1]**

[Total Marks: 7]